

SOPRON, Hungary

WMO: 128050

Lat: 47.6833N

Lon: 16.6000E

Elev: 784

StdP: 14.28

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.3	18.5	6.4	7.8	17.2	11.5	10.2	22.0	27.5	40.1	24.8	37.5	7.1	310	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	89.6	70.7	86.2	69.7	83.2	68.1	72.8	85.4	71.0	83.2	69.4	80.9	8.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.5	107.9	79.2	66.8	101.5	77.3	65.2	95.9	75.5	36.8	85.2	35.3	83.5	33.9	80.9	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.7	20.0	17.8	DB	9.1	94.0	6.0	3.8	4.8	96.8	1.3	99.0	-2.1	101.2	-6.4	104.0	(4)
			WB	8.0	75.5	5.8	1.8	3.8	76.8	0.4	77.9	-2.9	78.9	-7.2	80.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	32.4	36.1	43.3	52.6	60.4	67.0	70.4	69.4	60.7	51.7	42.5	33.8	(6)
(7)		DBStd	15.03	7.84	8.18	7.81	7.20	6.24	6.21	5.58	5.63	5.92	6.91	7.17	7.28	(7)
(8)		HDD50	2034	545	391	231	55	3	0	0	1	64	242	502		(8)
(9)		HDD65	5333	1009	808	672	374	170	50	17	21	156	414	676	966	(9)
(10)		CDD50	2683	1	3	24	134	324	511	631	602	321	115	16	1	(10)
(11)		CDD65	507	0	0	0	2	26	111	183	158	26	1	0	0	(11)
(12)		CDH74	4490	0	0	0	39	272	993	1622	1353	201	10	0	0	(12)
(13)		CDH80	1410	0	0	0	3	42	295	568	468	34	0	0	0	(13)
(14)	Wind	WSAvg	7.5	7.8	8.5	8.6	8.6	7.9	7.3	7.0	6.3	6.8	6.5	7.2	7.5	(14)
(15)	Precipitation	PrecAvg	26.2	1.1	1.1	1.6	1.8	2.9	3.4	3.1	3.0	2.6	2.1	2.0	1.5	(15)
(16)		PrecMax	33.0	2.5	3.0	3.5	3.5	6.9	7.5	7.8	6.3	6.7	4.8	4.4	3.1	(16)
(17)		PrecMin	19.7	0.2	0.1	0.1	0.0	0.7	1.0	0.0	0.5	0.5	0.2	0.2	0.1	(17)
(18)		PrecStd	3.8	0.6	0.8	0.9	0.9	1.5	1.6	1.8	1.5	1.5	1.2	1.0	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	56.1	61.2	69.4	78.8	84.2	91.3	93.3	93.4	84.4	75.2	65.1	56.3	(19)	
(20)		MCWB	47.2	49.1	53.3	61.1	65.6	71.3	71.6	71.4	67.5	62.7	56.7	49.3	(20)	
(21)		DB	50.7	55.6	64.6	73.8	80.5	86.9	89.6	89.5	79.2	70.2	60.0	50.6	(21)	
(22)		MCWB	44.0	46.7	51.4	58.1	64.4	70.5	70.9	70.3	66.2	60.6	54.1	45.5	(22)	
(23)		DB	46.8	51.9	60.0	69.8	77.0	83.3	86.2	85.4	75.7	66.4	55.9	46.8	(23)	
(24)		MCWB	41.4	44.9	49.3	56.0	63.2	68.3	69.6	69.5	64.1	58.6	51.4	42.2	(24)	
(25)		DB	43.5	48.0	55.7	66.0	73.3	80.0	82.8	81.7	72.0	62.3	52.5	43.9	(25)	
(26)		MCWB	39.6	42.2	47.2	54.0	60.7	66.6	68.3	67.9	62.1	56.6	48.8	40.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	48.2	50.1	54.9	62.0	68.7	74.9	74.6	74.9	69.6	64.3	58.1	49.8	(27)	
(28)		MCDWB	53.9	59.3	66.0	76.3	80.2	86.2	88.0	87.5	81.0	72.7	63.2	54.8	(28)	
(29)		WB	44.8	47.4	52.4	59.3	66.0	71.6	72.6	72.5	67.3	61.7	54.6	45.8	(29)	
(30)		MCDWB	49.6	54.6	62.5	71.4	77.3	83.7	85.5	85.3	77.2	68.5	59.2	49.9	(30)	
(31)		WB	42.0	45.1	50.2	57.0	64.0	69.6	71.0	70.6	65.2	59.5	51.9	42.9	(31)	
(32)		MCDWB	45.7	51.2	58.6	67.5	74.6	80.9	83.3	82.7	73.4	64.8	55.6	46.0	(32)	
(33)		WB	39.8	42.5	47.9	54.9	62.0	67.7	69.4	68.7	63.3	57.3	49.3	40.9	(33)	
(34)		MCDWB	43.4	47.5	54.7	64.4	71.4	78.3	81.0	79.5	70.3	61.8	51.9	43.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.6	13.2	16.2	18.9	18.9	19.1	19.5	19.5	17.3	15.9	11.2	10.0	(35)	
(36)		MCDBR	16.1	19.7	24.0	25.8	24.7	24.3	25.5	25.7	23.3	21.4	17.3	16.0	(36)	
(37)		MCWBR	11.2	12.5	13.6	13.2	12.2	11.1	11.1	11.2	11.8	12.8	11.6	11.5	(37)	
(38)		MCDBR	15.0	18.6	22.3	23.3	22.7	23.1	23.3	23.8	21.5	19.6	16.3	14.7	(38)	
(39)	Clear-Sky Solar Irradiance	MCWBR	10.9	12.3	13.1	12.3	11.8	11.0	10.8	11.3	11.3	12.1	11.4	11.0	(39)	
(40)		taub	0.314	0.338	0.376	0.419	0.417	0.421	0.427	0.428	0.402	0.373	0.345	0.314	(40)	
(41)		taud	2.398	2.361	2.285	2.188	2.242	2.264	2.249	2.260	2.305	2.362	2.396	2.424	(41)	
(42)		Ebn at Noon	241	259	266	264	269	268	264	258	253	242	226	226	(42)	
(43)	Edh at Noon	22	28	35	43	42	42	42	40	35	28	22	19	(43)		
(44)	All-Sky Solar Radiation	RadAvg	333	591	973	1437	1705	1854	1807	1606	1149	718	377	257	(44)	
(45)		RadStd	34	81	120	139	173	130	148	156	134	73	36	30	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (82 neighbors)	+1.03	N/A	N/A	+1.58	+0.86	+0.74	-211	-288	+166	+88	(47)

Nomenclature: See separate page